

TED[15] 6243 Reg. No.....
(REVISION-2015)

Signature.....

MODEL QUESTION PAPER

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/ MANAGEMENT/
COMMERCIAL PRACTICE**

MEDICAL IMAGING TECHNIQUES

[Time : 3 hours]

(Maximum marks : 100)

PART - A

(Maximum marks: 10)

Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define CT number.
2. Identify the range of frequency used in diagnostic ultrasound.
3. Define precession in MRI principle.
4. List 3 radioisotope used in nuclear medicine.
5. Identify the property of X-ray used in xenondetectors. (5x2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Explain the need for collimators and grids in X-ray machine.
2. Differentiate between radiography and fluoroscopy.
3. Write notes on speed of ultrasound in human body.
4. Explain interaction of ultrasound with matter.
5. Explain the process of resonance in MRI machine.
6. Compare CT and MRI machine with respect to application.
7. Describe the principle of operation of Gamma camera. (5x6=30)

PART - C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT - I

III.a. Describe the working of a CT machine with suitable block diagram. (8)

b. Explain the types of detectors used in CT machine. (7)

OR

IV.a. Write notes on

(i) Intra-aortic balloon pump.

(ii) Balloon angioplasty

(iii) X-ray films (5X3=15)

UNIT – II

V.a. Explain the principle of operation of Doppler machine. (6)

b. With the help of a block diagram, explain the working of B-mode scanner. (9)

OR

VI.a. Define piezoelectric effect. Describe the structure and function of ultrasound probes. (8)

b. List the applications of ultrasound scanners. (7)

UNIT – III

VII.a. Explain principle of operation and process of image formation in MRI machine (8)

b. Explain the need of gradient coils in MRI machine. List the types of gradient coils available in MRI machine. (7)

OR

VIII.a. Differentiate between T1 and T2 relaxation time. Explain its importance in image formation. (7)

b. With the help of a block diagram explain the working of an MRI machine. (8)

UNIT – IV

IX.a. Explain the working of a PET scanner with the help of a block diagram. (9)

b. Explain the types of detectors used in PET machine. (6)

OR

X.a. Describe the concept of nuclear imaging. Explain how radioisotopes are generated for this purpose. (8)

b. Explain the working of a pulse height analyzer in Gamma camera. (7)